IVONA WAYSIDE CLINIC

DHA HIE API Request Catalogue

Documented endpoints, request patterns, workflow sequences and source references

Scope
A working reference for the Ivona Wayside Clinic HMIS integration with the Digital Health Agency (DHA) Health Information Exchange (HIE) and SHA-related eClaims workflows. It separates current HIE documentation from older AfyaLink guide material and from details that still require confirmation in the authorized developer collection.

Prepared: 26 October 2026
Primary public source: DHA HIE documentation at hie-docs.dha.go.ke
Environment shown in current HIE docs: UAT base URL below; use only the environment assigned to your facility.

Important: this is a documented request catalogue, not proof that every endpoint is enabled for Ivona Wayside Clinic, that credentials have been issued, or that DHA certification is complete.

1. How to use this catalogue

This document is grounded in the uploaded Ivona API Integration Catalogue & Request Body Guide and the current public DHA HIE API documentation. Where the current online docs expose an endpoint but the full request schema is not visible in the retrieved page, the catalogue identifies the route and points to the official reference instead of inventing a payload.

Current HIE UAT base shown in DHA documentation

```
https://ilm-dev.dha.go.ke/uat-middleware/api/v1
```

For the HIE UAT routes below, append the documented path to this base. The current docs describe OAuth 2.0 and require an access token for authorized calls. Obtain the actual client credentials and facility-specific values through the DHA developer onboarding process.

Common headers

```
Accept: application/json
Authorization: Bearer <ACCESS_TOKEN>
Content-Type: application/json   # for JSON request bodies
X-Facility-Id: <FACILITY_FR_CODE>   # where required for service-account tokens
X-Facility-Id-Type: fr-code         # where required
```

The facility headers are documented for relevant HIE endpoints when a service-account token is used. Do not send made-up facility codes or tokens. Follow the exact auth scheme, scopes, and required headers for the assigned API client.

Request safety labels
• Read request: generally GET; may still reveal protected patient data.
• Action request: POST/PUT/PATCH/DELETE; may create a visit, send OTP, authorize consent, submit a claim, or change a record.
• Examples with dummy IDs are documentation examples only and must not be treated as real patients.

2. Authentication

2.1 Obtain an access token

POST /api/v1/tenants/token

Purpose: Obtain an access token for subsequent authorized HIE API requests.

Inputs / request body

```
{
  "client_id": "<ISSUED_CLIENT_ID>",
  "client_secret": "<ISSUED_CLIENT_SECRET>"
}
```

Documented response: The integration scenario documents an access_token response. Consult the current Authentication API reference for the complete grant, token fields, scopes, and error schema.

Notes: Do not put credentials in frontend code, screenshots, source control, or this PDF. Do not assume tokens for one service/environment authorize another.

Official reference: https://hie-docs.dha.go.ke/docs/authentication/getting-started/intro

3. Patient and facility registry requests

3.1 Search for a patient

GET /patients

Purpose: Search the HIE patient registry and retrieve the beneficiary Client Registry (CR) ID and demographics.

Inputs / request body

```
Query parameters documented by scenario pages: identification_number; identification_type.
```

Documented response: The integration scenario identifies the returned id as the beneficiary CR ID; store it as patient_id for downstream coverage/consent workflows.

Notes: Prefer ClientRegistry ID when already known; other identifier types may be used to resolve it. The exact allowed identification_type enum is documented on the eligibility reference. Use only an authorized test record in UAT.

Official reference: https://hie-docs.dha.go.ke/docs/scenarios/overview

3.2 Search facilities / bed occupancy

GET /facilities/{facilityCode}/beds/occupancy

Purpose: Retrieve facility bed occupancy data.

Inputs / request body

```
Path parameter: facilityCode (facility FR code).
```

Documented response: The docs describe bed_occupancy_rate, bp_level and name, with fields for total capacity and occupancy categories.

Notes: This route is explicitly documented in the current eClaims/eligibility reference. Do not confuse it with the older AfyaLink facility-search route.

Official reference: https://hie-docs.dha.go.ke/eclaims/eligibility

3.3 Older AfyaLink facility search route

GET /v2/facility-search?facility-fid={FACILITY_FID}

Purpose: Search facilities using the route recorded in the clinic's uploaded API guide.

Inputs / request body

```
Query parameter: facility-fid.
```

Documented response: Not reproduced here; confirm the response schema against the current authorized collection.

Notes: Legacy/guide-derived route, not confirmed as part of the current HIE UAT API. Keep separate from current HIE routes until DHA confirms the target environment.

Official reference: Clinic-uploaded PDF: 02_API_Integration_Catalogue_and_Request_Bodies.pdf

3.4 Older AfyaLink practitioner search route

GET /v1/practitioner-search?id={PRACTITIONER_UID}

Purpose: Search practitioner by UID using the route recorded in the clinic's uploaded API guide.

Inputs / request body

```
Query parameter: id.
```

Documented response: Not reproduced here; confirm the response schema against the current authorized collection.

Notes: Legacy/guide-derived route; do not assume it is enabled on the current HIE UAT host.

Official reference: Clinic-uploaded PDF: 02_API_Integration_Catalogue_and_Request_Bodies.pdf

4. SHA eligibility and benefit coverage

4.1 Check SHA eligibility

GET /patients/eligibility

Purpose: Check beneficiary eligibility and schemes under SHA.

Inputs / request body

```
Query parameters: identification_number; identification_type.
```

Documented response: Documented fields include memberCrNumber, schemes, statusCode, statusDesc, facilityBiometricsEnforced and whitelistedForOTP, plus patient demographics.

Notes: The docs say use ClientRegistry ID by default. If isAlive is false, the integration scenario says subsequent transactions for that patient must be blocked.

Official reference: https://hie-docs.dha.go.ke/eclaims/eligibility

4.2 Get patient benefits coverage

GET /patients/benefits

Purpose: Retrieve parent benefits the patient is entitled to.

Inputs / request body

```
Query parameter: patient_id (beneficiary CR ID).
```

Documented response: Paginated results containing parentBenefit and parentBenefitCode.

Notes: Use a CR ID returned by patient search; do not use a guessed identifier.

Official reference: https://hie-docs.dha.go.ke/eclaims/eligibility

4.3 Get patient sub-benefits coverage

GET /patients/sub-benefits

Purpose: List sub-benefits under parent benefits.

Inputs / request body

```
Required query: patient_id. Optional query fields documented: parent_benefit_code; fields.
```

Documented response: Sub-benefit records; the scenario flow uses sub_benefit_code for intervention lookup.

Official reference: https://hie-docs.dha.go.ke/eclaims/eligibility

4.4 Get intervention coverage

GET /patients/benefits/interventions

Purpose: List service interventions available to a patient.

Inputs / request body

```
Required query: patient_id. Optional filters include sub_benefit_code, parent_intervention, page, page_size, sub_interventio
```

Documented response: Intervention list. Key decision fields include paymentMechanism, needsPreauth, needsManualPreauthApproval, accessPoint and fund.

Notes: Use the returned intervention code and facility-level tariff values for the selected scenario; do not hard-code generic tariffs.

Official reference: https://hie-docs.dha.go.ke/eclaims/eligibility

4.5 Get benefit utilization balances

GET /patients/benefits/utilization

Purpose: Retrieve utilization and remaining limits for a specific intervention.

Inputs / request body

```
Required query parameters: patient_id; intervention_code.
```

Documented response: Fields documented include fundUtilizationLimit, individual/household limits, utilization days, next availability and computational details.

Official reference: https://hie-docs.dha.go.ke/eclaims/eligibility

4.6 Get POMSF balances

GET /patients/pomsf-balances

Purpose: Retrieve Public Officers Medical Scheme Fund balances and policy details.

Inputs / request body

```
Required query: patient_id. Optional documented fields: policy_year; principal_member_number.
```

Documented response: The reference lists member, family, policy, scheme and balance-related fields.

Notes: Use only where the patient/scheme is applicable; follow current endpoint schema.

Official reference: https://hie-docs.dha.go.ke/eclaims/eligibility

5. Consent, OTP and biometrics

The HIE integration scenarios require patient consent to start a visit. The patient contact/OTP path and biometric authorization path are distinct. These action requests must only be run with valid UAT credentials and an approved test patient.

5.1 Get beneficiary contacts

GET /patients/contacts

Purpose: Retrieve the patient's registered contact options for consent.

Inputs / request body

```
Use the patient identifier/query fields defined by the current Consent Services endpoint reference.
```

Documented response: Scenario documentation shows a contacts result with id, masked contactValue, contactType, isConfirmed and active.

Notes: Show masked contact numbers to the patient and confirm the selected contact before sending an OTP. The scenario overview identifies beneficiary_contact_id as the contact ID used by the OTP flow.

Official reference: https://hie-docs.dha.go.ke/docs/scenarios/overview

5.2 Send visit OTP

POST /claims/otp

Purpose: Send a one-time password to the beneficiary's registered contact to authorize starting a visit.

Inputs / request body

```
{
  "contact_id": "<CONTACT_ID_FROM_CONTACTS>",
  "intervention_codes": ["<VALID_SHA_INTERVENTION_CODE>"],
  "patient_id": "<BENEFICIARY_CR_ID>"
}
```

Documented response: The Consent Services reference documents a successful OTP operation; see live reference for exact response schema.

Notes: SIDE EFFECT: sends an OTP to the registered contact. Request-body field names can differ between documentation views (the send-OTP endpoint schema lists contact_id; scenario overview refers to beneficiary_contact_id). Follow the exact current request schema exposed by the authorized API collection before sending. Intervention codes must be valid for the visit.

Official reference: https://hie-docs.dha.go.ke/consent/send-otp

5.3 Create biometric authorization

POST /claims/authorize

Purpose: Create a pending consent authorization and initiate biometric matching through the returned iframe flow.

Inputs / request body

```
Biometric-specific fields are required. Use the exact BiometricsAuthorizationRequest schema in the current Consent Services A
```

Documented response: The scenario overview says authorization begins in PENDING; the response includes a token/consent identifier and a GUID used downstream. After successful matching, the authorization transitions to AUTHORIZED or AUTHORIZED_PENDING_VISIT depending on the workflow.

Notes: SIDE EFFECT: creates a consent authorization and may launch fingerprint matching. Do not call with a real patient unless explicitly authorized. See the official schema rather than relying on a hand-built payload.

Official reference: https://hie-docs.dha.go.ke/docs/scenarios/overview

5.4 Create visit / use consent

POST /claims/visit

Purpose: Start a patient visit using the consent result from the OTP or biometrics flow.

Inputs / request body

```
Use the exact visit request schema from the current eClaims API reference. The scenario docs state that the OTP is passed fo
```

Documented response: Successful visit creation returns visit/claim-related identifiers including the consent token used by downstream eClaims operations; verify exact response schema in the live reference.

Notes: SIDE EFFECT: creates a visit. Do not submit a guessed body or reuse consent data across visits.

Official reference: https://hie-docs.dha.go.ke/docs/scenarios/overview

6. Claims, discharge and preauthorization workflow

The current HIE scenario matrix chooses the workflow from returned intervention flags: fund, accessPoint (IP/OP), paymentMechanism, needsPreauth and needsManualPreauthApproval. The exact claim/preauthorization body schemas must be taken from the corresponding live endpoint reference and selected scenario.

6.1 Submit outpatient claim

POST /claims/submit

Purpose: Submit a claim for an outpatient visit.

Inputs / request body

```
Use the exact request body from the current Claims API schema and the scenario applicable to the patient's fund/intervention
```

Documented response: Claim submission/status fields are defined by the current Claims API reference.

Notes: SIDE EFFECT: submits a claim for processing. Verify tariff, diagnosis, intervention, consent and required attachments before submission.

Official reference: https://hie-docs.dha.go.ke/docs/scenarios/overview

6.2 Send discharge OTP

POST /claims/otp/discharge

Purpose: Send OTP to authorize discharge of an inpatient.

Inputs / request body

```
{
  "consent_token": "<CONSENT_TOKEN_FROM_VISIT>",
  "patient_id": "<BENEFICIARY_CR_ID>"
}
```

Documented response: The official endpoint page documents a standard success response with data and message.

Notes: SIDE EFFECT: sends an OTP. The example token and patient ID on the official site are dummy values, not credentials.

Official reference: https://hie-docs.dha.go.ke/consent/send-otp

6.3 Discharge inpatient and submit claim

POST /claims/discharge

Purpose: Discharge an inpatient; the scenario reference says this dispatches/submits the inpatient claim.

Inputs / request body

```
Use the current schema. OTP discharge uses the OTP field; biometric discharge uses auth_guid instead of otp, as described by
```

Documented response: See current Claims API response schema.

Notes: SIDE EFFECT: changes visit/discharge state and submits the claim. Confirm the patient, episode, authorization and billing details before use.

Official reference: https://hie-docs.dha.go.ke/docs/scenarios/overview

6.4 Close / discard a claim

POST /claims/close

Purpose: Discard/close a claim where the workflow permits.

Inputs / request body

```
Use the current Claims API request schema; no body is invented here.
```

Documented response: See current Claims API response schema.

Notes: SIDE EFFECT: may close or discard a claim. Confirm the exact business meaning and allowed state transitions before use.

Official reference: https://hie-docs.dha.go.ke/docs/scenarios/overview

6.5 Preauthorization endpoints

Confirm in current Claims API catalog **Exact path and body are scenario-dependent**

Purpose: The HIE documentation includes normal and elective preauthorization scenarios, but a single universal path/body should not be assumed.

Inputs / request body

```
Follow the specific scenario and live Claims API reference for create, update, cancellation and approval steps.
```

Documented response: Not summarized as a guessed route.

Notes: The scenario overview distinguishes inpatient/outpatient, FFS/per diem/capitation, and elective/manual preauthorization. Select the matching scenario before constructing requests.

Official reference: https://hie-docs.dha.go.ke/docs/scenarios/overview

6.6 Scenario selection guide

Scenario	Fund / care point	Payment / preauth	Claim dispatch
1. SHIF IP Per Diem	SHIF / Inpatient	Per diem; no preauth	At discharge
2. SHIF IP FFS normal preauth	SHIF / Inpatient	FFS; preauth, not elective	At discharge
3. SHIF IP FFS elective preauth	SHIF / Inpatient	FFS; elective preauth	At discharge
4. SHIF OP FFS elective preauth	SHIF / Outpatient	FFS; elective preauth	Submit
5. SHIF OP FFS normal preauth	SHIF / Outpatient	FFS; normal preauth	Submit
6. UHC OP capitation	UHC / Outpatient	Capitation	Submit
7. ECCIF emergency identified	ECCIF / OP or IP	Emergency; doctor consent	Submit within 24h
8. ECCIF emergency unidentified	ECCIF / OP or IP	Emergency; patient may not be identified	Submit within 24h

Source: DHA HIE Integration Scenarios overview. The scenario determines the correct sequence; this table is a routing summary, not a substitute for endpoint request schemas.

7. Shared Health Record (SHR) and legacy AfyaLink guide routes

The current DHA HIE API Catalog lists Shared Health Record (SHR) service APIs. The clinic's uploaded API guide also records older AfyaLink-style routes. These are deliberately separated: do not assume the older paths and FHIR base prefixes are valid on the current HIE UAT middleware host.

7.1 Older guide: submit FHIR bundle

POST /v1/shr-med/bundle

Purpose: Submit a FHIR JSON Bundle using the path recorded in the clinic's uploaded guide.

Inputs / request body

```
The uploaded guide contains a FHIR Bundle skeleton with Patient, Organization, Practitioner, Coverage and Claim resources, bu
```

Documented response: Use the current assigned API collection for response shape and claim processing states.

Notes: Legacy/guide-derived route; not confirmed as a current HIE UAT middleware endpoint. Do not submit the guide's skeleton to production.

Official reference: Clinic-uploaded PDF: 02_API_Integration_Catalogue_and_Request_Bodies.pdf

7.2 Older guide: query bundle status

GET /v1/shr-med/claim-status?bundle_id={BUNDLE_ID}

Purpose: Check status using a bundle ID, as documented in the clinic's uploaded guide.

Inputs / request body

```
Query parameter: bundle_id.
```

Documented response: Not reproduced; confirm with current API collection.

Notes: Legacy/guide-derived route. Confirm host and route before use.

Official reference: Clinic-uploaded PDF: 02_API_Integration_Catalogue_and_Request_Bodies.pdf

7.3 Older guide: query claim status

GET /v1/shr-med/claim-status?claim_id={CLAIM_ID}

Purpose: Check status using a claim ID, as documented in the clinic's uploaded guide.

Inputs / request body

```
Query parameter: claim_id.
```

Documented response: Not reproduced; confirm with current API collection.

Notes: Legacy/guide-derived route. Confirm host and route before use.

Official reference: Clinic-uploaded PDF: 02_API_Integration_Catalogue_and_Request_Bodies.pdf

8. Terminology service

The current DHA API Catalog lists a Terminology Service described as read-only APIs for resolving clinical terminology concepts and mappings (OCL-backed). The retrieved catalog summary did not expose the full endpoint paths or request schemas. Therefore this document does not invent terminology routes. Use the Terminology Service section of the official catalog.

Official DHA HIE API Catalog

9. Postman / PowerShell testing checklist

Before testing any endpoint, record its source, environment, method/path, authentication requirements, request schema, expected response, and whether it has side effects. Start with low-risk authenticated GET requests using an approved UAT test record.

Suggested order

1. Obtain UAT client credentials from the official DHA developer onboarding process.

2. Request an access token using the exact Authentication API contract and save it in Postman environment variables, not in the request body of unrelated endpoints.

3. Test a read-only registry request with an authorized UAT test identifier.

4. Run eligibility, then sub-benefit and intervention coverage calls using the returned CR ID.

5. Confirm the intervention's fund, access point, payment mechanism and preauthorization flags.

6. Review the patient contacts response and confirm the test contact before sending OTP.

7. Only then test OTP or biometric authorization; these create real workflow effects even in UAT.

8. Use the matching scenario to create a visit and test claim submission/discharge using approved dummy clinical/billing data.

9. Save sanitized request/response examples and correlation IDs; redact tokens, OTPs, national IDs, contact values and clinical data.

A safe route probe is not an API success test

The clinic's observed GET responses of HTTP 405 for /claims/otp and /claims/authorize are consistent with the documented POST method. OPTIONS returning 204 shows the server handled the OPTIONS request but, because no Allow header was returned, it does not independently enumerate supported methods. Neither result proves an authenticated POST will succeed.

10. Details that must come from the authorized DHA collection

• Exact token grant, credential placement, scopes/audience, expiry and refresh behavior.

• Complete query parameters and request/response schemas for patient contacts, visit, authorize, claims, preauthorization, discharge and close.

• Current biometrics authorization request schema and supported device/workstation requirements.

• Facility-specific FR code and any required facility headers.

• Exact claim-line schema, tariffs, code sets, attachments and scenario-specific constraints.

• Error codes, retry/idempotency rules, rate limits, callbacks and production go-live controls.

11. Official references

Current DHA HIE documentation consulted for this catalogue:

• **DHA HIE home and API services:** https://hie-docs.dha.go.ke/

• **API Catalog:** https://hie-docs.dha.go.ke/catalog

• **Integration Scenarios overview:** https://hie-docs.dha.go.ke/docs/scenarios/overview

• **Authentication introduction:** https://hie-docs.dha.go.ke/docs/authentication/getting-started/intro

• **Claims / Eligibility reference:** https://hie-docs.dha.go.ke/eclaims/eligibility

• **Consent Services introduction:** https://hie-docs.dha.go.ke/docs/consent/getting-started/intro

• **Send OTP endpoint reference:** https://hie-docs.dha.go.ke/consent/send-otp

Clinic-provided reference documents used:

• 02_API_Integration_Catalogue_and_Request_Bodies.pdf — identifies older AfyaLink-style routes and explicitly warns not to guess unverified token/eligibility/consent/preauthorization paths; it also labels the FHIR Bundle example incomplete.
• 04_DHA_Certification_Privacy_and_Security.pdf — used as context for privacy and authorization controls.
• 05_Environment_Configuration_Roles_and_Operations.pdf — used as context for environment separation and production safeguards.

Version note: the public documentation can change. Re-check each endpoint and schema in the authorized DHA developer collection before implementation or live testing. This document intentionally leaves unverified fields and paths unresolved rather than guessing.